



Cell division model

The first ever detailed model of stem cell division has been created and it is stunning! <https://digit.in/june19-54>



Preserve vaccines

Scientists have found a method to preserve vaccines for months without refrigeration. <https://digit.in/june19-55>

the sources of artificial light. They extend high into the atmosphere, and are located approximately in between the observer and the light sources. The light pillars are as such an optical phenomenon, and do not really have a physical location. If the ice crystals are floating in the air, the light pillars can be observed from light sources such as street lamps, even if they are located just a few meters away. While these may look like giant searchlights with lamps on the ground, they are not. Actually the ice crystals are reflecting the light downwards, towards the observer. As the light pillars are typically observed in the extreme latitudes, they are often photographed by aurora chasers. Since there are many kinds of aurorae, light pillars are often considered another exotic type of aurorae. This is however, not the case at all. Light pillars are caused by light being reflected from ground based sources, while aurorae are caused by high energy particles interacting with the upper atmosphere. Although they can appear together in some rare instances, aurorae have no relation to the light pillars.

SUN PILLARS

Sun pillars are a special kind of light pillars, where the light source is the sun. Again, the light is reflected from

millions of light crystals that are in the shape of hexagons or columns. However, these crystals can be located higher up in the atmosphere, and because of this reason, sun pillars can be seen from almost every location on Earth. Sun pillars are visible particularly during the times of sunrise or sunset. For Sun pillars to appear, the individual ice crystals have to be tilted sufficiently to reflect the sunlight back to the observer. A complete sun pillar shows a pillar of light extending upwards and downwards from the sun. Partial sun pillars can be either lower pillars or upper pillars. Depending on how high the sun is from the horizon, the length of the upper or lower pillars can change. The upper pillar reflects off the lower faces of crystals, while the lower pillar reflects off the upper faces of crystals. For this reason, the lower pillar often extends below the horizon, while the upper pillar truncates as it appears to reach towards the sky. As the sun sets, the upper pillars elongate and become brighter, a process that can continue even after the sun has disappeared behind the horizon. Of the two, the lower pillars are easier to spot. The best places to observe lower pillars are from high altitude locations, such as the tops of mountains.

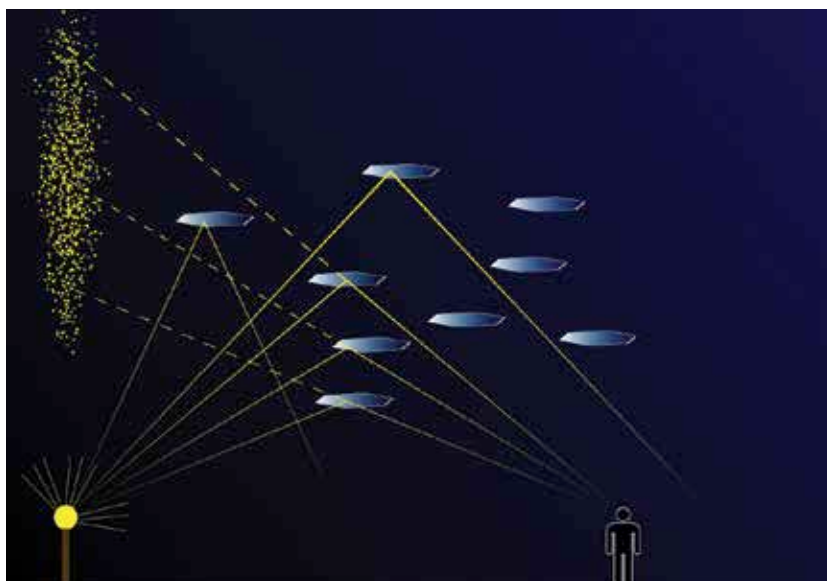
The shape of the crystals causes them to act like parachutes, and they



Credit: Mugregg

A Sun Pillar

reflect the light from the sun as they drift downwards slowly. In the right conditions, when the skies are clear and the moon is bright, and the crystals are present in the high altitudes as well, the moon can cause light pillars as well. The light pillars are shaped according to the phase of the moon. If it is a full moon, the light pillars are also uniform. However, crescent moons cause light pillars where the light suddenly falls off on one side. To cause this phenomenon, light is travelling from the sun, bouncing off the moon in the shape of Earth's shadow on it, reflecting off the high altitude ice crystals, before finally creating the image of a pillar in the eyes of an observer. In very rare circumstances, light pillars can even be seen around planets, particularly Venus. These light pillars are far tinier than the pillars caused by the sun, the moon or ground-based light sources. They are localised around the planet, and appear as tiny extensions on either side. The visibility can be significantly improved with a good pair of binoculars as well. The light pillars around Venus are best observed after seeing a light pillar from the sun, because the presence of high altitude ice crystals is confirmed. While other planets do produce light pillars as well, if they are bright and the skies are clear, they might be too small to be visible to the naked eye. ❗



How light pillars are formed